Introduction to Module 4: Data Structures Foundations

Design and Analysis of Algorithms
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Why Study Data Structures

- Critical for performance of a wide range of algorithms and their implementations.
- Important to understand what's "under the hood", say when using built-in data structures.
- They help illustrate elegant ideas that can generalize to design and analysis of other algorithms.

3 Core Data Structures to Remember

- Binary Search Trees (this module)
- Hash Tables (later in course)
- Binary Heaps (this module)

- These give efficient implementations for a wide range of abstract “types” of data structures: sets, maps, priority queues, sequences, etc.

Module Contents

- Binary Heaps
- Binary Search Trees
  - The basics
  - Using them to store sequential data
  - Keeping them balanced
- Example of using these structures in algorithmic problem-solving via "sweep line" / "sort and scan" methods.
- Coding discussion: Build a balanced binary search tree
- Enrichment (optional): Applications of Treaps / Cartesian trees (these are structures that hybridize heaps and binary search trees)

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